

Algorithm for the construction of Rectilinear Steiner Minimum Tree by identifying the clusters of points

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Abstract— The proposed algorithm constructs Rectilinear Steiner Minimum Tree (RSMT) by applying clustering algorithm to identify the required number of cluster of points and then applies a strategy to construct RSMT for each of the cluster. The clusters are then interconnected to get the final RSMT. The constructed RSMT produces improvement over Rectilinear Minimum Spanning Tree in terms of wire length reduction. RSMT are widely used in the global routing phase of VLSI design and wire length reduction.

Keywords—Minimum Spanning Tree; Rectilinear Steiner Minimum Tree; Rectilinear Minimum Spanning Tree; VLSI Design

I. INTRODUCTION

Routing in VLSI design is the process of connecting the cells that have been assigned position as a result of placement. Routing is generally performed in two different stages. The first stage, called the global routing, will identify the wiring channels through which connections can run. The second stage, called detailed routing, fixes the exact paths that the wire has to take.

Given a set of n points $P = \{p_1, p_2, p_3, \dots, p_n\}$, Rectilinear Steiner Minimum Tree (RSMT) is a tree constructed over P with some additional points called Steiner Points. Steiner points are added to reduce the length and to connect in rectilinear manner. The distance between two points is measured in the rectilinear metric. The rectilinear distance between a pair of points $p_i = (x_i, y_i)$ and $p_j = (x_j, y_j)$ is equal to $|x_i - x_j| + |y_i - y_j|$.

Hanan[1] was the first person to consider the rectilinear version of Steiner Tree. He gave exact solutions for $n \leq 5$ and also showed that the possible candidate Steiner points (CSp) lie on Hanan Grid (Fig. 2). Garey and Johnson [2] showed that the problem of constructing RSMT is NP-Complete and Hwang [3] proved that the ratio of cost of Rectilinear Minimum Spanning Tree (RMST) to the cost of RSMT is $\leq 3/2$. Fig. 1 shows the RSMT constructed over points P_1 to P_5 using Steiner Points S_1, S_2 and S_3 .

Many approximation algorithms exist for the construction of RSMT. The popular ones are: Khang and Robins [4] presented an $O(n^3)$ Iterated 1-Steiner (IIS) and $O(n^4 \log n)$ Batched Iterated 1-Steiner algorithm which iteratively adds one and batch of Steiner points respectively and constructs

RMST. Borah, Owens and Irwin [5] proposed an $O(n^2)$ edge-based heuristic algorithm which first constructs MST and incrementally improves it by connecting a point to the enclosing rectangular layout of the edge visible to point. Zhou [6] proposed a $O(n \log n)$ Rectilinear Spanning graph (RSG) algorithm based on Borah et al edge based heuristic [5] on the Zhou et al spanning graph algorithm [7]. Khang, Mandoiu and Zelikovsky [8] proposed an $O(n \log^2 n)$ batched version of greedy triple contraction algorithm [9] - Batched Greedy Algorithm [BGA], where a batch of triples (optimal full Steiner tree for a set of three points where all the points are leaves) are added in each iteration to construct RSMT. Wong, Yiu-Chung and Chu [10] proposed an $O(n \log n)$ fast look-up table based algorithm. Table for computing RSMT for $n \leq 9$ are pre-computed. For $n > 9$ an optimal net breaking algorithm is used. The various algorithms for the construction of RSMT have been surveyed in [11].

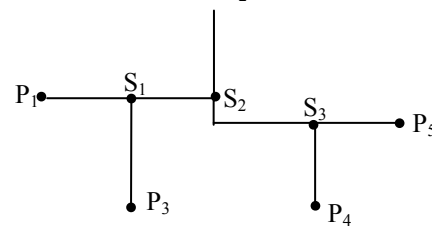


Fig. 1 Rectilinear Steiner Minimum Tree over points P

II. BASIC DEFINITION

A. Rectilinear Minimum Spanning Tree

Minimum Spanning Tree (MST) is a minimum length tree over a given set of points. Rectilinear Minimum Spanning Tree (RMST) is a MST connected in rectilinear manner by restricting the edges to horizontal and vertical lines. Fig. 2 shows MST over points P_1 to P_6 .

B. Hanan Grid

Hanan Grid is a grid induced by drawing horizontal and vertical lines through each and every given point. Fig. 2 shows the Hanan grid induced by the points P_1 to P_6 .

C. Hanan Points

The points obtained at the intersection of horizontal and vertical lines drawn through the given set of points P are

called as Hanan points [1]. For a given point set P with n points, there are $O(n^2)$ Hanan points which are the possible candidate Steiner points.

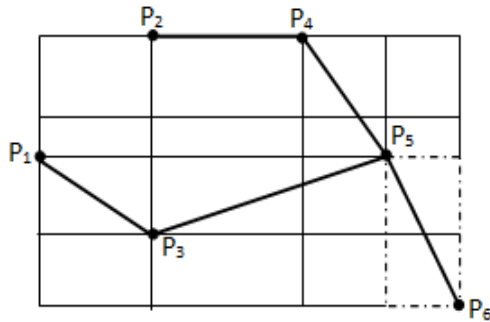


Fig. 2 MST over points P_1 - P_6 and Hanan Grid

III. ALGORITHM

The algorithm for the construction of RSMT is carried out in three steps.

1. Identification of k (user input variable) clusters.
2. Construction of RSMT for each cluster.
3. Connecting the clusters to identify the final RSMT.

A. Identification of k clusters

Given a set of n points, the first step is to identify k clusters using a clustering algorithm. The clustering algorithm selected is k -means clustering algorithm since it is proved to be fast, robust, easier to understand, relatively efficient and produces good result when the given point set is distinct. The k -means clustering algorithm identifies the k -clusters by applying the following procedure:-

1. ' k ' cluster centers (centroids) are randomly selected.
2. The distance between each points and cluster centers are calculated.
3. Each point is assigned to the cluster which has a shortest distance between point and cluster center.
4. When all the points are assigned to clusters, the cluster centers are recalculated.
5. Step 2, 3 and 4 are repeated until the centers no longer move.

B. Construction of RSMT for each cluster

Once the required numbers of clusters are formed, Rectilinear Steiner Minimum Tree is constructed for each of the cluster. For the P_i number of points in C_i cluster, Minimum Spanning Tree (MST) is constructed using prim's algorithm. For each and every edge of MST two possible candidate Steiner points (CSp_1 and CSp_2) are identified. The candidate Steiner points lie on the hanan grid and out of these two candidate Steiner points, the point that has shortest distance to rest of the points in the cluster is selected. Similarly, Steiner points are selected for the remaining edges of the MST and the

RSMT is constructed by including these identified Steiner points. The methodology is repeated for all the clusters and RSMT for each of the cluster is constructed.

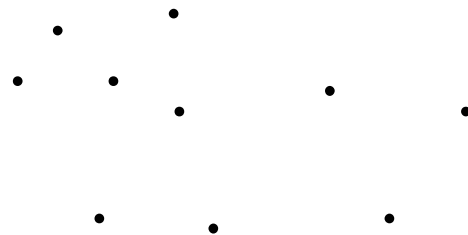
C. Connecting the clusters to identify the final RMT

After constructing RSMT for each of the k -clusters, the clusters have to be finally connected to obtain the final RSMT over the given set of points. To connect the clusters, the centroid of each cluster is identified. Each of the centroid is considered as a point and MST of k -centroid points is constructed. According to the MST, if centroid of cluster₁ is to be connected to centroid of cluster₃ then the closest points of cluster₁ and cluster₃ are connected by selecting the Steiner point that has shortest distance to the centroids of remaining clusters. Similarly, the remaining clusters are connected to obtain the final Rectilinear Steiner Minimum Tree.

The complete algorithm is as follows:-

1. Compute k -clusters over the point set P using k -means clustering algorithm. Let the clusters be $C_1, C_2, C_3, \dots, C_k$.
2. For each cluster C_i with m points compute RSMT as follows:
 - a) Compute MST for the m points in cluster C_i using Prim's algorithm.
 - b) For each edge of $e=(p,q)$ of MST, two possible candidates Steiner points are CSp_1 and CSp_2 . Candidate Steiner point that has shortest distance to all the points in C_i is selected as Steiner point and the Steiner points are added to the point set.
3. Compute MST for the centroids of k -clusters using Prim's algorithm. According to the MST, if cluster i (C_i) is to be connected to cluster j (C_j) then P_i is connected to P_j such that $P_i \in C_i, P_j \in C_j$ and the length of the edge P_iP_j is shortest among all edges connecting cluster i and cluster j . The edge P_iP_j is connected using the Steiner point having the shortest distance to the centroids of remaining clusters.

For example, consider the 12 points as shown in fig 3. If the number of clusters selected is 3, then the three clusters constructed using step 1 of algorithm is as shown in Fig. 4. Fig. 5 shows the RSMT constructed for each of the cluster by applying step 2. Final RSMT over the given 12 points is obtained by applying step 3 (Fig. 6).



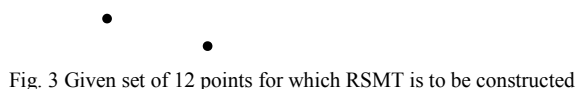


Fig. 3 Given set of 12 points for which RSMT is to be constructed

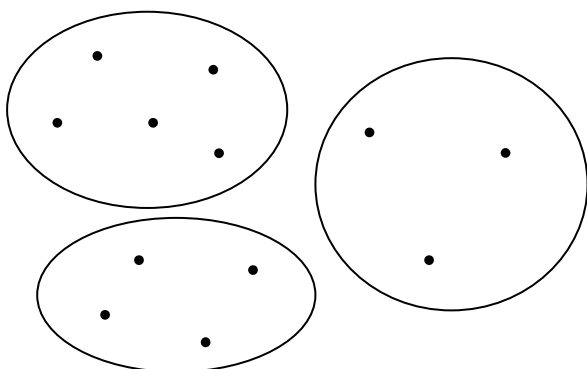


Fig. 4 Point set P divided into three clusters

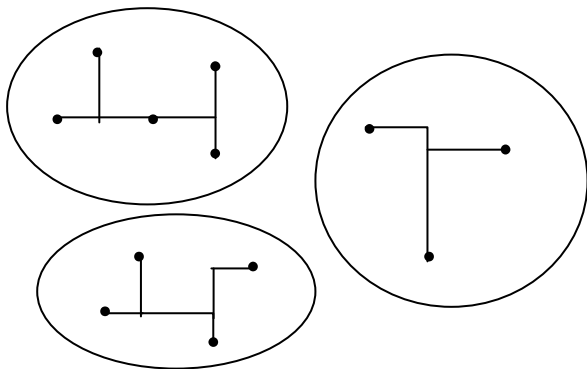


Fig. 5 RSMT constructed for the points in each of the cluster

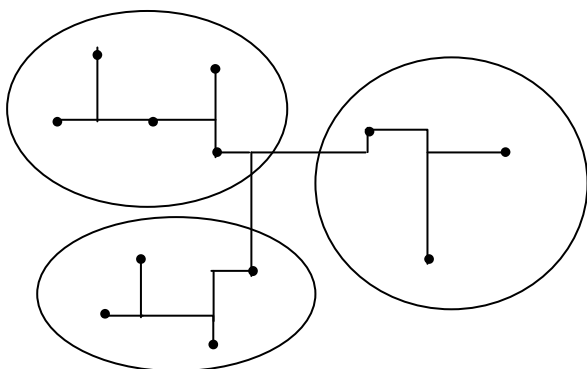


Fig. 6 Final RSMT for the given point set constructed by connecting the clusters

IV. IMPLEMENTATION AND RESULTS

The entire algorithm is implemented in C language. For the computation of k -clusters, $O(n \log n)$ implementation of k -means algorithm is used. Prim's algorithm has a time complexity of $O(n^2)$. The time complexity of the proposed algorithm is $O(km^2)$ where k is the number of clusters and m is the number of points in each cluster.

Table 1 shows the results obtained for different number of points and different number of clusters. Table also shows the comparison of the RSMT and RMST with respect to total length reduction (ratio of length of RMST to the length of RSMT).

TABLE I
COMPARISON OF RECTILINEAR STEINER MINIMUM TREE AND RECTILINEAR MINIMUM SPANNING TREE

# of Points	# of clusters	Ratio of length of RMST to the length of RSMT
20	3	1.076
40	4	1.071
60	4	1.061
100	4	1.059
200	4	1.053
400	4	1.04
500	5	1.04
800	5	1.04
1000	6	1.04

V. CONCLUSION

We have proposed an algorithm that efficiently constructs Rectilinear Steiner Minimum Tree for the given point set P. Clustering algorithm has been used for the first time in constructing the RSMT with the advantage of the problem being divided into k sub-problems that can be dealt individually. The strategy used for constructing RSMT for each cluster works efficiently. As shown in Table I, the proposed algorithm provides good improvement over RMST in terms of length reduction. Future efforts would be directed towards reducing the time complexity of the algorithm and parallel computing since each cluster is independent of other clusters.

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